science	professor	title	subject	affiliations	toys	reasoning	www
Mathematics	Michael Friedlander	Professor; Director IAM	Convex optimization, ML algorithms, signal processing, sparse recovery	IAM Director; SIAM Fellow (2024)	SPGL1 software, optimization libraries	Bridges math+CS; directly relevant to AI foundations	www
Mathematics	Leah Edelstein-Keshet	Emeritus Professor	Mathematical cell biology, cell motility, pattern formation, ecological modeling	RSC Fellow; SIAM von Neumann Prize (2022)	Mathematical modeling tools	2025 CRM-Fields-PIMS Prize; textbook Mathematical Models in Biology	www
Mathematics	Ahmet Alacaoglu	Assistant Professor	Optimization algorithms for ML, stochastic methods, min-max games	AIM-SI (CAIDA); CS associate member	ARC Sockeye	New hire 2024; ICML/NeurIPS publications; AI+math bridge	www
Mathematics	Eric Cytrynbaum	Professor	Mathematical biology, cell biophysics, microtubule dynamics, pattern formation		Experimental + computational modeling	MATH 560 graduate course in mathematical biology	www
Mathematics	Philip Loewen	Professor	Optimal control, nonsmooth analysis, RL applications, variational analysis		Optimization software	Control theory foundations for RL	www
Mathematics	Trevor Campbell	Associate Professor	Scalable Bayesian inference, Bayesian nonparametrics, streaming data		Bayesian inference tools	NeurIPS/ICML publications; coresets for scalable inference	www
Computing	Mark Schmidt	Associate Professor	ML optimization, large-scale learning, structured prediction, medical image analysis	CAIDA	ARC Sockeye GPU cluster, ML/DL frameworks	Core ML optimization; directly relevant to AI foundations	www
Computing	Kevin Leyton-Brown	Professor; Co-Director CAIDA	ML + game theory, algorithmic mechanism design, electronic markets	CAIDA (Co-Director); ACM Fellow; AAAI Fellow	ARC Sockeye, optimization/simulation software	CAIDA leadership; AI decision-making; h-index 54	www
Computing	Frank Wood	Associate Professor; Director PLAI	Probabilistic programming, autonomous driving, computational neuroscience, robotics, NLP	PLAI; CIFAR AI Chair (Amii)	Open-source probabilistic programming tools	Probabilistic ML for science; Anglican language	www
Computing	Leonid Sigal	Associate Professor; CRC II	Computer vision, object recognition, human pose estimation, action recognition	CV Lab; CIFAR AI Chair (Vector); CRC II	Motion capture systems, 3D vision lab	CV Lab co-founder; h-index 59; vision for science	www
Computing	Kwang Moo Yi	Assistant Professor	Visual geometry, 3D CV, autonomous vehicles, drones, robotics	CV Lab; CAIDA; ICICS	Computer vision hardware	Area chair CVPR/ICCV; 3D vision for autonomous systems	www
Computing	Michiel van de Panne	Professor	Physics-based character animation, RL for control, robot locomotion	MOCCA Lab; SIGGRAPH Academy	Physics simulation, motion capture	2022 SIGGRAPH Achievement Award; RL for motion	www
Computing	Jeff Clune	Associate Professor	Deep RL, neuroevolution, genetic algorithms for neural networks	CIFAR AI Chair (Vector); DeepMind advisor	GPU clusters, evolutionary computing	Pioneer deep neuroevolution; on leave 2026	www
Computing	Giuseppe Carenini	Professor	NLP, discourse parsing, summarization, sentiment analysis, visual text analytics	UBC NLP Group (Co-Director)	NLP infrastructure	NLP for psychiatric clinical notes; interdisciplinary	www
Computing	Raymond Ng	Professor; Director DSI	Data mining, text mining, health informatics, genomics data analysis	Data Science Institute (Director); Language Sciences	Data mining platforms	SIGKDD/SIGMOD best papers; top Academic Data Leader	www
Computing	Geoff Pleiss	Assistant Professor	Neural network uncertainty, Bayesian optimization, Gaussian processes, scientific discovery	AIM-SI (CAIDA); CIFAR AI Chair (Amii); Vector	GPyTorch library	AI Methods for Scientific Impact cluster; bridges ML and science	www
Computing	David Poole	Professor Emeritus	Knowledge representation, probabilistic reasoning, computational logic	LCI; CAIAC Lifetime Achievement	AILog2 reasoning system	Foundational AI textbooks; reasoning under uncertainty	www
Computing	Ian Mitchell	Professor	Control theory, verification, cyber-physical systems, level set methods	Scientific Computing Lab; DFP	Level Set Methods Toolbox	Safety of human-in-loop systems; assistive technology	www

Computing	Rafeef Garbi	Professor; Director BiSICL	AI for medical imaging, deep learning, computer vision, 3D image processing	BMIAI group; Bioinformatics	Medical imaging infrastructure	Deep learning for cancer CT, pediatric screening; AI+biology bridge	www
Computing	Artem Cherkasov	Professor; CRC I	Computer-aided drug discovery, AI/ML drug screening, Deep Docking	Vancouver Prostate Centre; CRC I	$24M ADDUCT infrastructure for CADD	Deep Docking screens billions of molecules; AI+chemistry bridge	www
Physics	Andrea Damascelli	Professor; Scientific Director QMI	Quantum materials, ARPES, time/spin-resolved spectroscopy, unconventional superconductors	Blusson QMI (Director); Max Planck-UBC-UTokyo	Advanced ARPES/REXS systems	Killam/Brockhouse/Steacie Prizes; Royal Society of Canada	www
Physics	Alison Lister	Professor; CRC	Top quarks, BSM physics, dark matter, deep learning in particle physics	ATLAS/CERN; CRC; 2025 Breakthrough Prize	ATLAS detector, deep learning infrastructure	ML for particle classification; USRA project on deep learning at ATLAS	www
Physics	Colin Gay	Professor	Higgs boson, top quark measurements, ATLAS DAQ system	ATLAS/CERN; 2025 Breakthrough Prize	ATLAS detector, custom DAQ electronics	Led Canadian team for Higgs discovery	www
Physics	Jess McIver	Associate Professor; CRC II	Gravitational waves, LIGO detector characterization, multi-messenger astronomy	LIGO Deputy Spokesperson; CRC II	LIGO data, Python/C++ analysis	GW170817 team; 2025 UBC McDowell Award; USRA on LISA analysis	www
Physics	Sarah Burke	Associate Professor; CRC II	Scanning probe microscopy, organic electronics, 2D materials, topological properties	CRC II Nanoscience	Ultra-high vacuum STM/AFM at 5K	Atomic-scale imaging; quantum materials	www
Physics	Joshua Folk	Professor	Quantum transport, mesoscopic physics, quantum Hall effect, van der Waals heterostructures	Quantum Devices Group; QMI	Nanoscale devices, ultra-low temp systems	Nature Physics, Science publications; quantum thermodynamics	www
Physics	Kirk Madison	Professor	Cold atoms, quantum gases, quantum sensors, vacuum metrology	Madison Quantum Gas Lab	Ultracold atom systems, quantum sensors	Tunable Rb-Li interactions; quantum vacuum metrology	www
Physics	Kris Sigurdson	Associate Professor	Dark matter, particle cosmology, dark energy, inflation, cosmic 21-cm	Former NASA Hubble Fellow; IAS Member	Computational cosmology tools	Hylogenesis theory; dark matter-antimatter asymmetry	www
Physics	Thomas White	Professor	HED physics, extreme states, National Ignition Facility, European XFEL		NIF, European XFEL, LCLS access	World-class pulsed laser and x-ray facilities	www
Physics	Daniel Trugman	Associate Professor	Seismology, earthquake rupture, ML in seismology, induced seismicity	Seismological Lab	Seismic waveform analysis, GrowClust software	ML in seismology; directly relevant to AI+earth science	www
Chemistry	Curtis Berlinguette	Professor; Distinguished Scholar	AI + lab automation for materials discovery, solar cells, CO2 conversion	Project Ada; $200M Acceleration Consortium	Automated high-throughput screening labs	Self-driving lab pioneer; AI+chemistry flagship	www
Chemistry	Jason Hein	Associate Professor	Automated reaction analysis, robotics + organic synthesis, self-driving labs	Project Ada (Co-Founder)	Robotic reaction profiling, in-situ IR/LC-MS	AI-driven synthesis; real-time reaction monitoring	www
Chemistry	Roman Krems	Professor; Distinguished Scholar	ML for quantum chemistry, quantum ML, few/many-body quantum systems	APS Fellow; RSC Member; Kavli Fellow	Quantum computing resources	ML+quantum chemistry; directly relevant to AI+science	www
Chemistry	Yan Alexander Wang	Professor	Quantum chemistry, DFT, computational materials science, catalysis		Computational chemistry infrastructure	Linear-scaling first-principles methods	www
Chemistry	Corey Stephenson	Professor; CERC	Visible light photoredox catalysis, electrochemistry, natural products synthesis	Canada Excellence Research Chair	Flow chemistry, photoredox equipment	Pioneer photoredox catalysis; new hire 2024	www
Chemistry	Corinna Schindler	Professor	Iron-catalyzed synthesis, carbonyl-olefin metathesis, earth-abundant metal catalysis	ACS Award in Pure Chemistry; Sloan Fellow	Synthesis labs	ACS Pure Chemistry Award; new hire 2024	www
Chemistry	Katherine Ryan	Professor; CRC I	Biosynthetic enzymes, antibiotic/ anticancer pathways, protein engineering, X-ray crystallography	CRC I Biosynthetic Enzymes	X-ray crystallography, protein engineering	CSC Young Investigator; McDowell Award	www

Chemistry	David Perrin	Professor	Bioorganic chemistry, DNA aptamers, PET imaging, radiopharmaceuticals		PET imaging, radiolabeling equipment	Teva Award CSC; baited-SELEX for cancer targeting	www
Chemistry	Glenn Sammis	Professor	Radical fluorination, heterocycle synthesis, natural product total synthesis, electrochemistry		Synthesis labs, electrochemistry equipment	Radical chemistry for pharmaceuticals; Harvard PhD (Jacobsen)	www
Chemistry	Gregory Dake	Associate Professor	Total synthesis of natural products, transition metal catalysis, ligand design	Killam Teaching Prize	Synthesis labs	Associate Head for Undergrads; strong teaching focus	www
Chemistry	Eva Nichols	Assistant Professor; CRC II	Electrocatalysis for CO2 reduction, bio-inspired catalyst design	CRC II Molecular Electrocatalysts	Electrochemistry equipment	Artificial photosynthesis; UC Berkeley PhD	www
Chemistry	Mark MacLachlan	Professor; CRC	Supramolecular chemistry, structural color, nanotubes, sensing	CRC Supramolecular Materials; New Dean of Science (2025)	Materials characterization	Steacie Prize; new Dean of Science	www
Chemistry	Raymond Andersen	Professor	Marine natural products, drug discovery, biosynthetic studies	CIC Medal	Natural products isolation equipment	FDA-approved drug from his research; Farnsworth Award	www
Biology	Inanc Birol	Scientist & Professor	Bioinformatics, genome assembly, sequence mapping, visualization	Michael Smith Genome Sciences Centre; CS cross-appt	HPC, ABySS 2.0, ORCA platform	ABySS: 1500+ citations; central to genomics computing	www
Biology	Paul Pavlidis	Professor	Neuroinformatics, bioinformatics, computational biology, neuropsychiatric disorders	Pavlidis Lab; Michael Smith Labs; DMCBH	Gemma gene expression platform	Pioneer neuroinformatics; integrates genomics+neuroscience	www
Biology	Michael Kobor	Associate Professor; CRC I	Epigenetics, social epigenetics, gene expression, environmental influences on health	CRC I Social Epigenetics; Edwin Leong Chair	Epigenetics analysis platforms	Fetal alcohol, asthma, COPD epigenetics; worldwide collaborations	www
Biology	Mark Cembrowski	Associate Professor	Computational neuroscience, molecular neuroscience, memory circuits, RNAseq	DMCBH; Allen Institute; HHMI Janelia	Patch-clamp, computational modeling	BS Math (UBC), PhD Applied Math; bridges math+neuroscience	www
Biology	Loren Rieseberg	Professor; CRC; Director BRC	Plant evolutionary genomics, bioinformatics, adaptation, speciation	Biodiversity Research Centre (Director); CRC	Greenhouse, genomic tools	#1 in Canada for Ecology/ Evolution; sunflower genomics	www
Biology	Jörg Bohlmann	Professor	Plant genomics, terpenoid biochemistry, chemical ecology, conifer defense	Treenomix; Michael Smith Labs	Genomics platforms	Canada first large-scale forestry genomics (Treenomix)	www
Biology	Terry Snutch	Professor; Research Chair	Molecular neurobiology, calcium channels, neuronal signaling, neurological disease	Michael Smith Labs; Brain Research Centre	Electrophysiology, molecular genetics	Discoveries basis for FDA-approved therapeutics (Neuromed)	www
Biology	Dolph Schluter	Professor	Evolutionary ecology, speciation, adaptive radiation, conservation genetics	Biodiversity Research Centre	Field research, genomics	Distinguished expert in evolutionary biology	www
Biology	Darren Irwin	Professor	Evolutionary biology, ornithology, conservation genomics, species identification	Biodiversity Research Centre	Genomic analysis, vocal analysis software	Integrative approach: genomics + computation + field	www
Biology	Sally Otto	Professor	Population genetics, mathematical modeling, evolutionary theory, genetic diversity	BRC; Fellow Royal Society (UK)	Mathematical modeling tools	Mathematical biology; Fellow of Royal Society	www
Biology	Laura Parfrey	Associate Professor	Microbial ecology, microbiome, protists, high-throughput sequencing	Parfrey Lab	HTS platforms, phylogenetic tools	Eukaryotic microbe diversity; community characterization	www
Biology	Chris Harley	Professor	Coastal marine ecology, climate change impacts, ocean acidification, intertidal	Institute for the Oceans and Fisheries	Field equipment, marine stations	Climate-marine interaction; Bamfield station access	www
Biology	Marco Marra	UBC Killam Professor	Cancer genomics, disease genomics, epigenetics	Genome Sciences Centre (leadership)	Sequencing platforms, HPC	Killam Professor; genomics leadership	www

Astronomy	Douglas Scott	Professor	CMB, observational cosmology, astro-statistics, ML methods in astronomy	Planck/Herschel/SCUBA-2	Computational cosmology	Hosts AI for Physics colloquia; USRA on ML in astronomy	www
Astronomy	Mark Halpern	Professor	CMB, observational cosmology, star formation, BLAST telescope	BLAST; SCUBA-2; JCMT	Submillimeter telescopes	Actively recruits students; balloon-borne astronomy	www
Astronomy	Michelle Kunimoto	Assistant Professor	Exoplanet detection and characterization, TESS mission, transit searches	TESS mission team	TESS data, Python analysis tools	Discovered 100+ confirmed exoplanets; USRA on JWST/ ARIEL	www
Astronomy	Ingrid Stairs	Professor	Radio pulsars, FRBs, gravitational waves via NANOGrav, CHIME telescope	NANOGrav; CHIME; 2025 RSC Fellow	CHIME radio telescope	CHIME discovering pulsars/FRBs; Rutherford Medal	www
Astronomy	Aaron Boley	Associate Professor; CRC	Planet formation, protoplanetary disks, exoplanet dynamics, space sustainability	CRC Planetary Astronomy; Outer Space Institute	Thunderbird South Observatory (Chile)	OrbDot Python package; former NASA Sagan Fellow	www
Astronomy	Brett Gladman	Professor; CRC	Solar system dynamics, Kuiper Belt, asteroid discovery, LSST analysis	CRC Planetary Astronomy; CFEPS; OSSOS	Telescope time, LSST data access	USRA on LSST data; asteroid 7638 named after him	www
Astronomy	Allison Man	Assistant Professor	Galaxy evolution, star formation, supermassive black holes, JWST data analysis	BCKDF funding for JWST infrastructure	JWST data, Gemini Observatory	USRA on galaxy evolution with JWST; Python required	www
Astronomy	Gary Hinshaw	Professor	CMB anisotropy, cosmological parameters, COBE-DMR, WMAP satellite	Former NASA Goddard; COBE/WMAP	Radio observatory development	Building new radio observatories at UBC	www
Astronomy	Ludovic Van Waerbeke	Professor	Dark matter mapping, gravitational lensing, dark energy, structure formation	2015 Bessel Award	Weak lensing analysis tools	Maps dark matter through gravitational lensing	www
Earth Science	Lindsey Heagy	Assistant Professor; Director GIF	ML for geophysical data, inverse theory, EM imaging, mineral exploration	GIF Director; CAIDA member	SimPEG software, GeoSci.xyz	Neural networks for geophysical inversion; AI+earth science flagship	www
Earth Science	Valentina Radić	Associate Professor	Glaciology, climate change, sea level rise, ML in geosciences		Glacier measurement systems	First global glacier projections; teaches ML for geosciences since 2013	www
Earth Science	Rachel White	Assistant Professor	Atmospheric dynamics, extreme weather, paleoclimate, climate modeling	Climate Dynamics Group	Climate modeling infrastructure	Physics background (Cambridge); climate dynamics	www
Earth Science	Philippe Tortell	Professor; Dept Head	Ocean productivity, biogeochemical cycling, climate-active gases, mass spectrometry		Sea-going mass spectrometry	Pioneer marine gas measurement	www
Earth Science	Susan Allen	Professor	Physical oceanography, coastal dynamics, microplastic transport, canyon flows		SalishSeaCast ocean model	Numerical ocean modeling; 30+ years at UBC	www
Earth Science	Mark Jellinek	Professor	Volcanology, geodynamics, planetary science, geological fluid mechanics	Geophysical Fluids Lab	Experimental and computational dynamics	Volcanic-climate-tectonic coupling	www
Earth Science	Erik Eberhardt	Professor; Dept Head	Rock mechanics, landslide mechanics, numerical modeling, geotechnical monitoring	Rio Tinto Centre; ICaRN leader	Geotechnical monitoring equipment	Franklin Award for rock mechanics; industry partnerships	www
Earth Science	William Hsieh	Professor Emeritus	ML for environmental sciences (textbook author), atmospheric/ocean ML		ML for Environmental Sciences textbook	First textbook on ML for environmental science (Cambridge 2009)	www

science	name	type	people	overview	toys	access	reasoning	www
Mathematics	Institute of Applied Mathematics	Institute	Michael Friedlander (Director); 160+ faculty	6 clusters: math biology, nonlinear dynamics, scientific computing, image/data, optimization, fluid mechanics	Mathematical modeling tools, ARC Sockeye	Open to UBC researchers	Cross-departmental applied math hub	www
Computing	CAIDA	Research Centre	Kevin Leyton-Brown (Co-Dir), Alan Mackworth (Co-Dir), 100+ faculty	Centre for AI Decision-making and Action; UBC primary AI org spanning 30 departments; AIM-SI cluster for scientific AI	ARC Sockeye GPU cluster, optimization/simulation software	Open to UBC researchers; seminar series public	Central AI hub; AIM-SI directly targets AI for science	www
Computing	Computer Vision Lab	Lab	Leonid Sigal, Kwang Moo Yi, Evan Shelhamer, Helge Rhodin	Image/video understanding, 3D vision, pose estimation, vision-language models; created RoboCup and SIFT features	Motion capture systems, 3D vision hardware, professional CV lab	Open to grad and undergrad through USRA	One of most influential CV labs globally	www
Computing	PLAI	Lab	Frank Wood (Director)	Pacific Laboratory for AI; probabilistic programming, autonomous systems, computational neuroscience, robotics, NLP	Open-source probabilistic programming tools	Open to grad and undergrad through USRA	Applied probabilistic ML; cross-disciplinary partnerships	www
Computing	Lab for Computational Intelligence	Lab	David Poole, Alan Mackworth	Knowledge representation, reasoning, robotics, constraint-based systems; real-time visually-based control	Sensor-based robotic systems, mobile robot teams	Open to grad and undergrad through USRA	Foundational AI textbooks; AILog2 reasoning system	www
Computing	MOCCA Lab	Lab	Michiel van de Panne (Director)	Physics-based character animation, RL for control, motion modeling; DeepMimic	Physics simulation, motion capture equipment	Open to grad and undergrad through USRA	SIGGRAPH Achievement Award; RL for physical control	www
Computing	UBC NLP Group	Lab	Giuseppe Carenini (Co-Dir), Vered Shwartz	NLP, discourse parsing, summarization, commonsense reasoning, health NLP	NLP computing infrastructure	Open to grad and undergrad through USRA	Psychiatric NLP; commonsense reasoning	www
Computing	Data Science Institute	Centre	Raymond Ng (Director)	Central hub for data science research; computing resources for ML/data-intensive research	Data mining platforms, computing resources	Open to UBC researchers	University-wide data science support	www
Computing	ARC Sockeye HPC	Infrastructure	UBC Advanced Research Computing	General-purpose HPC: 400+ CPU nodes (15,872 cores), 50 GPU nodes (200 NVIDIA V100), InfiniBand EDR, 1.5 PB Lustre + 5 PB	200 NVIDIA Tesla V100 GPUs, 15872 CPU cores, 1.5 PB Lustre, 5 PB	Free to all UBC researchers; application-based	Primary compute; comparable to UNR Pronghorn	www
Physics	Blusson Quantum Matter Institute	Research Centre	Andrea Damascelli (Scientific Dir), Joshua Folk, Sarah Burke, 50+ scientists	First GREx Institute at UBC; quantum materials identification, classification, exploration; IBM Quantum System One workshops	Advanced ARPES/REXS, STM/AFM at 5K, quantum devices, cleanroom nanofab	Through QMI; open to faculty/students	World-class quantum materials; Max Planck partnership	www
Physics	TRIUMF	National Facility	UBC-hosted consortium of 21 universities; 500+ scientists	Canada particle accelerator centre; worlds largest normal-conducting cyclotron; proton/pion/muon/radioactive ion beams	520 MeV cyclotron, ISAC beams, detector systems	~30 Co-op/USRA positions annually	Major national facility on UBC campus; unique access	www
Physics	ATLAS/CERN UBC Group	Collaboration	Alison Lister, Colin Gay	Higgs boson, top quarks, BSM physics, deep learning for particle classification; ATLAS DAQ system	ATLAS detector access, deep learning infrastructure	USRA projects available (Python)	Deep learning at ATLAS; 2025 Breakthrough Prize	www
Physics	Gravitational Wave Group	Lab	Jess McIver (lead)	LIGO detector characterization, multi-messenger astronomy, gravitational wave analysis; LISA development	LIGO data access, Python/C++ analysis tools	USRA projects available	LIGO Deputy Spokesperson; GW170817 team	www
Physics	PHAS Cleanrooms	Facility	Department	Film preparation, patterning (sub-micron), crystal growth, sample cutting/polishing	Cleanroom equipment, film deposition, patterning systems	Department members	Recently commissioned; quantum device fabrication	www
Physics	PHAS Machine Shop	Facility	Department technical services	Woodworking, metal machining, sheet metal, soldering; experienced machinist on-site	Metal/wood machining, soldering, hand tools	Students, staff, faculty; 8am-4pm M-F	Support for experimental physics projects	www
Chemistry	Project Ada / Self-Driving Lab	Lab	Curtis Berlinguette, Jason Hein	AI-driven materials discovery; automated synthesis + ML for clean energy materials; $200M Acceleration Consortium	Automated high-throughput screening, robotic reaction profiling, in-situ IR/LC-MS	Open to grad and undergrad	AI+chemistry flagship; self-driving lab pioneer	www
Chemistry	Krems Group	Lab	Roman Krems	ML for quantum chemistry, quantum ML, dynamics of quantum systems	Quantum computing resources	Open to grad and undergrad	ML+quantum chemistry; APS/RSC Fellow	www
Chemistry	NMR Facility	Core Facility	Dr. Zhicheng (Paul) Xia	7 spectrometers: Bruker AV300, AV400inv, AV400dir, AV600 cryoprobe, AV300 SIF, Solid State AV400, AV III HD 400	3x Bruker 400 MHz, 1x Bruker 600 MHz cryoprobe, solid-state NMR	Drop-in after training (2x 1.5hr sessions)	NMR is essential for organic chemistry; 7 spectrometers	www

Chemistry	Mass Spec/ Microanalysis Facility	Core Facility	Dr. Yun Ling	10 instruments: Waters LCMS, Bruker HCT/ MALDI-TOF, Agilent GCMS/QTOF, Thermo LTQ-Orbitrap/Quantis, Flash 2000 EA	Agilent QTOF (40K res), Thermo Orbitrap (100K res), MALDI-TOF, elemental	Self-service (trained) and fee-based	Comprehensive mass spec; ESI/APCI/MALDI/ EI/CI/FD/FI	www
Chemistry	X-Ray Crystallography Facility	Core Facility	Dr. Brian Patrick	5 diffractometers: Bruker D8 Venture (Ag/Cu), Apex DUO, Panalytical Empyrean 3, D8-Advance, D8-Discover (-180 to +800C)	Bruker D8 Venture dual microsource, powder diffraction, variable temp	Full service; crystallographer as co-author	Single crystal + powder XRD; variable temp to 800C	www
Chemistry	Shared Instrument Facility	Core Facility	Ben Herring (Director)	12 instruments: AFM, DSC, CD, fluorescence, FTIR, GC-MS, HPLC, HPLC-MS, NMR 300, polarimeter, UV-Vis, TGA	AFM, DSC, CD spectrometer, FTIR-ATR, GC-MS, HPLC-MS, NMR,	Training required; online booking; NMR drop-in	Broad access to characterization tools	www
Chemistry	Stephenson Group	Lab	Corey Stephenson	Visible light photoredox catalysis, flow chemistry, natural products synthesis	Flow chemistry equipment, photoredox reactors	Open to grad and undergrad	CERC; photoredox pioneer; new hire 2024	www
Chemistry	Schindler Group	Lab	Corinna Schindler	Iron-catalyzed synthesis, carbonyl-olefin metathesis	Organic synthesis equipment	Open to grad and undergrad	ACS Pure Chemistry Award; new hire 2024	www
Biology	Genome Sciences Centre	Centre	Marco Marra (Dir), Steven Jones, Inanc Birol	Cancer/rare disease/infectious disease genomics; sequencing platforms; ABySS genome assembly	Sequencing platforms, HPC, bioinformatics pipelines	Collaborative; open to UBC researchers	Canada genomics flagship; ABySS tool central	www
Biology	Michael Smith Laboratories	Centre	Peter Zandstra (Director); 300+ researchers; 20+ groups	Interdisciplinary biotechnology: health and environmental challenges; genomics core, proteomics core	NextSeq, NovaSeq X, liquid handlers, proteomics MS	Open to UBC researchers	Cross-faculty interdisciplinary hub	www
Biology	Biodiversity Research Centre	Centre	Loren Rieseberg (Director); 50+ scientists	#1 in Canada for Ecology/Evolution; genes to ecosystems; Beaty Museum (2M+ specimens)	Genomics tools, field equipment, 2M+ museum specimens	Open to UBC researchers and students	#1 Canada ecology/ evolution; 5-8th globally	www
Biology	UBC Bioimaging Facility	Core Facility		Most comprehensive biological imaging in Western Canada: 3 TEMs, 2 SEMs, confocal/ multi-photon, cryo-fixation, CLEM	FEI Tecnai G2, Hitachi H7600/S2600, Zeiss XB350, high-pressure	Fee-for-service; training provided	Western Canadas top biological imaging	www
Biology	HRMEM Cryo-EM Facility	Core Facility		Structural biology: single-particle cryo-EM, cryo-tomography; COVID spike protein analysis	Glacios, Titan Krios (energy-filter), Aquilos 2 Cryo-FIB (only one in	Member Canadian Cryo-EM Nexus; consultation	Only Aquilos 2 in Canada; structural biology flagship	www
Biology	Sequencing + Bioinformatics Consortium	Core Facility		NGS: library prep, sequencing, bioinformatics analysis; RNA-Seq, ChIP-Seq, WGS	Illumina MiSeq (2x300 PE, 15Gb), NextSeq2000	Fee-for-service; training available	Central sequencing for all UBC	www
Biology	ubcFLOW Cytometry	Core Facility		Cell sorting and analysis: 6 sorters + 7 analyzers; 20+ years operation	Astrios, BD Aria, BD Influx, Cytek Aurora sorters	Fee: $80/hr UBC, $150/hr industry	Comprehensive flow cytometry	www
Biology	FINDER CL3 Facility	Core Facility		Containment Level 3: bacterial/viral infectious disease; 6700 sq ft with micro, proteomics, imaging, flow suites	CL3 suites: microbiology, proteomics, imaging, flow cytometry	Through Life Sciences Institute	Biosafety level 3; rare capability	www
Biology	Institute for the Oceans and Fisheries	Centre	Chris Harley + 50 experts	AERL facility (55,000 sq ft, $8.9M); sustainable oceans, fisheries, coastal ecosystems	AERL lab facility, marine research equipment	Open to UBC researchers	Major marine science hub	www
Biology	Bamfield Marine Sciences Centre	Field Station	UBC consortium (UBC/UVic/ SFU/UAlberta/UCalgary)	Marine biology, oceanography, coastal science; west coast Vancouver Island	Marine research facilities	Seasonal; consortium access	World-class marine field station	www
Astronomy	Thunderbird South Observatory	Observatory	Aaron Boley, Peter Hickson, Harvey Richer	0.5m telescope in Chilean Andes (~5590 ft); remotely operated by UBC students; deep-sky imaging	0.5m telescope, CCD cameras, remote operation software	Remote access for UBC students	Southern hemisphere; student-operated; comparable to GBO	www
Astronomy	UBC Campus Observatory	Observatory	Department	42-cm Cassegrain/coudé telescope with spectrograph and 2048x2048 CCD; student training and stellar research	42-cm telescope, coudé spectrograph, 2048x2048 CCD	Students and public	On-campus hands-on astronomy	www
Astronomy	CHIME Radio Telescope	Collaboration	Ingrid Stairs + UBC team	Canadian Hydrogen Intensity Mapping Experiment; discovering pulsars and FRBs; built by UBC	Radio telescope array	Through CHIME collaboration	UBC-built; NANOGrav gravitational wave evidence	www
Earth Science	Geophysical Inversion Facility	Lab	Lindsey Heagy (Director)	ML for geophysical data; SimPEG open-source inversion; GeoSci.xyz education platform	SimPEG software, computational infrastructure, ARC Sockeye	Open; SimPEG is open-source	AI+geophysics flagship; open-source tools	www

Earth Science	PCIGR	Core Facility		Pacific Centre for Isotopic and Geochemical Research: electron microprobe, SEM, XRD	JEOL JXA-iHP200F EPMA, Hitachi SU3900 SEM, Bruker D8 XRD	UBC researchers	Geochemical analysis hub	www
Earth Science	UBC-Teck Field Station	Field Station		36-hectare geological field station near Oliver, Okanagan; 3-season operation	Field training facilities	UBC students (field school)	Hands-on geological fieldwork	www
Earth Science	Nevada Seismological Lab equivalent	Lab	Daniel Trugman + department	Earthquake Research Facility: two shake tables for seismic testing since 2003	Shake tables, seismic monitoring	Department members	Seismic testing; BC Ministry of Education research	www
External	DeLaMare-equivalent: HATCH Makerspace	Makerspace	ICICS	3D printing, laser cutting, water-jet cutting, metal sintering, lathes, drill presses	3D printers, 3D scanner, laser cutter, water-jet, metal bending, lathes	UBC community	Prototyping and fabrication hub	www
External	AMPEL Nanofabrication Facility	Cleanroom		~50 m² cleanroom (ISO-5 and ISO-6); photolithography, thin film deposition, patterning	NXQ4006 mask-aligner, Heidelberg MLA150, deposition systems	Fee-based; training required	Sub-micron patterning; nanofabrication	www
External	UBC MRI Research Centre	Core Facility		Canadas first Philips 3.0T MR7700 (2023); PET/MR; fMRI, DTI, TMS	Philips 3.0T MR7700, GE SIGNA 3.0T PET/MR	Research projects	Neuroimaging; Canadas first MR7700	www
External	Pacific Museum of Earth	Museum		Geology, paleontology, mineralogy; public museum with educational programming	Earth/planetary science exhibits and collections	Public; educational programs	Earth science outreach; K-12 programs	www

science	lab	technology	toys	www
Mathematics	Institute of Applied Mathematics	Applied Math, Optimization, Modeling	160+ faculty, 6 research clusters, mathematical modeling tools, ARC Sockeye access	www
Computing	Advanced Research Computing Sockeye HPC	HPC, AI/ML, GPU Computing	200 NVIDIA V100 GPUs, 15,872 CPU cores, 1.5 PB Lustre filesystem, 5 PB Chinook object storage, InfiniBand EDR	www
Computing	Data Science Institute	Data Mining, Health Informatics, ML	Computing resources, data mining platforms, training programs	www
Computing	Computer Vision Lab	Computer Vision, 3D Reconstruction, Pose	Motion capture systems, 3D vision hardware, professional CV lab equipment	www
Physics	TRIUMF	Particle Accelerator, Nuclear Physics	520 MeV cyclotron (worlds largest normal-conducting), ISAC beamlines, detector systems, ~30 Co-op/USRA annually	www
Physics	Blusson Quantum Matter Institute	Quantum Materials, Spectroscopy, Devices	Advanced ARPES/REXS, ultra-high vacuum STM/AFM at 5K, quantum device cleanroom, IBM Quantum workshops	www
Physics	PHAS Cleanrooms & Machine Shop	Nanofabrication, Machining, Prototyping	ISO cleanrooms, film deposition, sub-micron patterning, metal/wood machining, soldering stations	www
Physics	AMPEL Nanofabrication Facility	Photolithography, Thin Film, Patterning	NXQ4006 mask-aligner, Heidelberg MLA150 maskless, ISO-5/ISO-6 cleanrooms (~50 m^2)	www
Physics	UBC MRI Research Centre	MRI, fMRI, PET/MR, Neuroimaging	Philips 3.0T MR7700 (Canadas first), GE SIGNA 3.0T PET/MR, multiband DTI, interleaved TMS/fMRI	www
Physics	HATCH Makerspace (ICICS)	Prototyping, 3D Printing, Fabrication	3D printers/scanner, laser cutter, water-jet, metal sintering printer, lathes, drill presses	www
Chemistry	NMR Facility	NMR Spectroscopy	7 spectrometers: Bruker AV300/400/600 MHz cryoprobe, solid-state AV400, auto-switchable probes	www
Chemistry	Mass Spectrometry Facility	Mass Spectrometry, Elemental Analysis	Agilent QTOF (40K res), Thermo LTQ-Orbitrap (100K res), Bruker MALDI-TOF, Thermo Quantis TSQ, Flash 2000 EA	www
Chemistry	X-Ray Crystallography Facility	X-Ray Diffraction, Crystal Structure	Bruker D8 Venture (dual Ag/Cu), Apex DUO, Panalytical Empyrean 3, D8-Advance/ Discover (-180 to +800°C)	www
Chemistry	Shared Instrument Facility	General Characterization	AFM, DSC, CD, fluorescence, FTIR-ATR, GC-MS, HPLC-MS, NMR 300 MHz, polarimeter, UV-Vis, TGA	www
Chemistry	Project Ada / Self-Driving Lab	AI-Driven Synthesis, Lab Automation	Automated high-throughput screening, robotic reaction profiling, in-situ IR/LC-MS, ML integration	www
Biology	Genome Sciences Centre	Genomics, Sequencing, Bioinformatics	Sequencing platforms, HPC, ABySS assembly software, bioinformatics pipelines	www
Biology	Michael Smith Laboratories	Biotechnology, Genomics, Proteomics	Illumina NextSeq + NovaSeq X, high-throughput liquid handlers, proteomics MS, 300+ researchers	www
Biology	Sequencing + Bioinformatics Consortium	NGS, RNA-Seq, ChIP-Seq, WGS	Illumina MiSeq (2×300 PE, 15 Gb), NextSeq2000, bioinformatics consultation	www
Biology	UBC Bioimaging Facility	Electron Microscopy, Confocal, Cryo-EM	3 TEMs (FEI Tecnai, Hitachi), 2 SEMs, confocal/multi-photon, high-pressure freezer, Vitrobot, CLEM	www
Biology	HRMEM Cryo-EM Facility	Cryo-Electron Microscopy	Glacios screening, Titan Krios (energy-filter), Aquilos 2 Cryo-FIB (only one in Canada)	www
Biology	ubcFLOW Cytometry	Flow Cytometry, Cell Sorting	6 sorters (Astrios, BD Aria, BD Influx, Cytek Aurora), 7 analyzers, multi-color	www
Biology	FINDER CL3 Facility	Containment Level 3 Research	6,700 sq ft CL3: microbiology, proteomics, imaging, flow cytometry suites	www
Biology	Biodiversity Research Centre / Beaty Museum	Ecology, Evolution, Collections	2M+ specimens, genomics tools, field equipment, herbarium, entomology, marine invertebrates	www
Biology	Institute for the Oceans and Fisheries	Marine Science, Fisheries	AERL facility (55,000 sq ft), marine research equipment, field stations	www
Biology	Bamfield Marine Sciences Centre	Marine Biology, Coastal Research	Marine research facilities, west coast Vancouver Island, consortium (5 universities)	www
Astronomy	Thunderbird South Observatory	Telescope, CCD Imaging, Remote	0.5m telescope in Chilean Andes, CCD cameras, remote operation software, student-operated	www
Astronomy	UBC Campus Observatory	Telescope, Spectroscopy	42-cm Cassegrain/coudé, coudé spectrograph, 2048×2048 CCD camera	www
Astronomy	CHIME Radio Telescope	Radio Astronomy, FRBs, Pulsars	Radio telescope array (UBC-built), pulsar timing, fast radio burst detection	www